Identification in Local Projections

Assumptions, estimands, derivations and literature

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1 The basic answer

The usual approach is to identify plausibly exogenous variation in a policy variable or structural shock, then estimate its relationship with the outcome separately at each horizon. A local projection (LP) is the regression method; the identifying design supplies the causal interpretation. A typical specification is

$$Y_{t+h} = \alpha_h + \beta_h D_t + \gamma_h' W_{t-1} + u_{t+h}, \qquad h = 0, \dots, H. \tag{1}$$

Here D_t is the treatment or policy variable, W_{t-1} is a chosen information set, and Y_{t+h} is the outcome. The label "shock" is reserved below for an innovation with an explicit identifying argument. An ordinary policy change need not qualify.

Route	Source of identifying variation	Main burden of argument
Conditional exogeneity	Variation left after conditioning on information	Relevant confounders are controlled, with adequate adjustment
Recursive restrictions	Innovations ordered by within-period timing	The assumed contemporaneous exclusions are credible
Observed shock	Narrative or high-frequency shock measure	Measurement, timing and exclusion of other news
External instrument	A proxy correlated with the target shock	Relevance and exclusion at the required horizons

These routes overlap. A narrative measure may be used directly or as an instrument; a recursively identified innovation can enter an LP. Sign restrictions, long-run restrictions and other structural designs also exist. The four routes above are a practical teaching classification, not an exhaustive taxonomy.

Teaching point. Always separate three questions: Which causal response is desired? What assumptions identify it? Which estimator and uncertainty calculation are appropriate?

These notes develop explicit sufficient conditions and pedagogical examples. They are not a claim that all cited papers impose the same model. Sections 11–13 connect the review to the course manuscript; the final sections give a design checklist, worked exercises and an annotated reading guide.

Reading: Jordà (2005); Jordà and Taylor (2025).

2 Define the intervention before the regression

Let $\mathcal{F}_{t-1}$ describe information before the date-t intervention. Write $Y_{t+h}(d)$ for the potential outcome when the date-t treatment is set to d, keeping the realised past fixed and allowing subsequent variables to evolve under a specified continuation rule. This notation suppresses the past treatment history and future disturbances; it does not erase them.

Three distinct targets are useful:

$$\tau_h(d, \delta \mid z) = \mathbb{E}[Y_{t+h}(d+\delta) - Y_{t+h}(d) \mid Z_{t-1} = z], \tag{2}$$

$$\tau_h^{\text{shift}}(\delta \mid z) = \mathbb{E}[Y_{t+h}(D_t+\delta) - Y_{t+h}(D_t) \mid Z_{t-1} = z], \tag{3}$$

$$\mu_h(d \mid z) = \frac{\partial}{\partial d}\mathbb{E}[Y_{t+h}(d) \mid Z_{t-1} = z]. \tag{4}$$

Equation (2) compares two fixed treatment values. Equation (3) shifts each realised treatment by the same amount. Equation (4) is a local derivative. Without restrictions on the response function these are different objects. A one-standard-deviation coefficient is a change of units; it does not establish that a finite change is proportional to a derivative.

2.1 A shock intervention differs from fixing a policy path

Suppose $R_t = a(\mathcal{F}_{t-1}) + X_t$, where R_t is the policy rate and X_t a policy innovation. Changing X_t today while retaining the future policy rule allows future rates to respond to the resulting economic conditions. This is the usual total impulse response. Holding $R_{t+1}, \ldots, R_{t+H}$ fixed is a different intervention and generally needs a different design. Including those future rates as ordinary controls does not automatically identify that counterfactual.

2.2 Consistency, timing and support

Consistency requires $Y_{t+h} = Y_{t+h}(D_t)$ under the observed regime. The initial state must be measured before the relevant news arrives. If a policy is announced today and implemented next quarter, the implementation date can be too late to define the surprise.

For nonparametric identification, treatment values being compared must be in the conditional support at the histories over which the response is averaged. A positive conditional variance alone is insufficient for an arbitrary finite shift. If treatment has bounded support, $D_t + \delta$ may leave it for some observations. One can restrict the target population, choose a support-preserving intervention, or state the extrapolation assumptions explicitly. Continuous-treatment conditional means are identified almost everywhere; smoothness is needed to discuss values or derivatives at individual points.

The outcome also determines the target: a level Y_{t+h}, a change $Y_{t+h} - Y_{t-1}$, and a sum of flows $\sum_{j=0}^{h} Y_{t+j}$ are not interchangeable. The response of a pre-treatment baseline is zero under no anticipation, but estimation samples and precision can still differ.

Reading: Rambachan and Shephard (2025 revision) provide a potential-outcome framework for dynamic causal interpretation.

3 Conditional exogeneity and adjustment for information

A useful sufficient restriction is conditional mean exogeneity: for each admissible d,

$$\mathbb{E}[Y_{t+h}(d) \mid D_t, W_{t-1}] = \mathbb{E}[Y_{t+h}(d) \mid W_{t-1}]. \tag{5}$$

Together with consistency and support, this identifies the conditional response surface

$$\mathbb{E}[Y_{t+h}(d) \mid W_{t-1} = w] = \mathbb{E}[Y_{t+h} \mid D_t = d, W_{t-1} = w].$$

Full conditional independence is stronger than this mean restriction. Merely adding a few lags does not establish either one. The information set must remove the dependence between treatment assignment and the relevant counterfactual outcome.

3.1 When a single LP coefficient is causal

For clarity, assume a constant affine causal response at horizon h:

$$\mathbb{E}[Y_{t+h}(d) \mid W] = m_h(W) + \theta_h d. \tag{6}$$

Then (5) implies $Y_{t+h} = m_h(W) + \theta_h D_t + u_{t+h}$ with $\mathbb{E}[u_{t+h} \mid D_t, W] = 0$. Define

$$p(W) = \mathbb{E}[D_t \mid W], \qquad V_t = D_t - p(W), \qquad \mathbb{E}[V_t^2] > 0.$$

Because $\mathbb{E}[V_t \mid W] = 0$ and $\mathbb{E}[V_t u_{t+h}] = 0$,

$$\theta_h = \frac{\mathbb{E}[V_t Y_{t+h}]}{\mathbb{E}[V_t D_t]} = \frac{\mathbb{E}[V_t Y_{t+h}]}{\mathbb{E}[V_t^2]}. \tag{7}$$

Thus a finite treatment increase of δ changes the conditional mean by $\delta\theta_h$. This interpretation used both exogeneity and the constant affine restriction.

3.2 Linear controls are a separate specification choice

Ordinary least squares with a finite vector W residualises using the best *linear* projection $P_W D_t$, not necessarily $p(W)$. Let $V_t^L = D_t - P_W D_t$, including an intercept in W. Under the model above,

$$\beta_h^L = \theta_h + \frac{\mathbb{E}[V_t^L m_h(W)]}{\mathbb{E}[(V_t^L)^2]}.$$

The extra term vanishes if, for example, $m_h(W)$ is in the linear span of the controls, or $p(W)$ is correctly represented by that span. Flexible adjustment can relax those linearity requirements. It cannot compensate for omitted information that is needed for (5).

Teaching point. "Conditional exogeneity given the past" and "OLS with four lags" are different claims. Explain why the selected controls and their functional form implement the desired adjustment.

Reading: Jordà and Taylor (2025), identification discussion; Lloyd and Manuel (2024), comparison of shock construction and direct control adjustment. The algebra here is a simple partially linear illustration.

4 Heterogeneity, orthogonalisation and what residuals do

4.1 Conditional heterogeneity creates variance weights

Replace (6) by $\mathbb{E}[Y_{t+h}(d) \mid W] = m_h(W) + \theta_h(W)d$. Even with exact residualisation,

$$\frac{\mathbb{E}[V_t Y_{t+h}]}{\mathbb{E}[V_t^2]} = \frac{\mathbb{E}[\mathrm{Var}(D_t \mid W)\theta_h(W)]}{\mathbb{E}[\mathrm{Var}(D_t \mid W)]}. \tag{8}$$

This is a variance-weighted average slope. It need not be the unweighted average $\mathbb{E}[\theta_h(W)]$. For binary treatment, the weights are $p(W)\{1-p(W)\}$. The weights describe the population parameter; increasing the sample size does not change it into a different target.

4.2 The exact FWL equivalence has conditions

In a common sample with the same controls and ordinary least squares, let $\widehat{V} = M_W D$, where M_W projects off W. The coefficient on D in the regression of Y on (D, W) equals $(\widehat{V}'Y)/(\widehat{V}'\widehat{V})$. It also equals the slope in the regression of $M_W Y$ on $\widehat{V}$. This is the Frisch–Waugh–Lovell identity.

The identity need not survive changing the sample, controls, weights, nonlinear estimator or regularisation between stages. With horizon-specific samples, a residual constructed on the full sample may not be orthogonal to the controls in every estimation sample. Moreover, the residual from regressing raw Y only on $\widehat{V}$ differs from the residual after controlling for W, so its naive standard error need not match the one-step calculation. Lloyd and Manuel (2024) analyse why these distinctions matter in macroeconomic applications.

4.3 Unpredictability is not causal exogeneity

An innovation satisfies $\mathbb{E}[X_t \mid \mathcal{F}_{t-1}] = 0$. This removes predictable components, not contemporaneous confounding. Consider independent sequences U_t, η_t with zero means and unit variances:

$$X_t = U_t + \eta_t, \qquad Y_t(x) = \theta x + \lambda U_t.$$

Then X_t is independent over time and unpredictable from the past, but

$$\frac{\mathrm{Cov}(X_t, Y_t)}{\mathrm{Var}(X_t)} = \theta + \frac{\lambda}{2}.$$

The true effect of increasing x by one is θ. A contemporaneous omitted disturbance produces the discrepancy despite a perfectly unpredictable regressor. This is a teaching example, not an empirical claim about any particular shock series.

Teaching point. Residualisation establishes an orthogonality property relative to the fitted controls. It does not by itself establish independence from potential outcomes or prove that the remaining variation is the structural shock of interest.

5 Recursive identification and the LP–VAR connection

Consider a linear structural moving-average system with orthogonal structural shocks. A reduced-form innovation vector can be written

$$u_t = B\varepsilon_t, \qquad \mathbb{E}[\varepsilon_t \varepsilon_t'] = I.$$

A recursive identification makes B lower triangular for a particular ordering. In a two-variable example,

$$u_{y,t} = b_{11}\varepsilon_{1t}, \qquad u_{d,t} = b_{21}\varepsilon_{1t} + b_{22}\varepsilon_{2t}.$$

The identifying restriction is that the policy shock ε_{2t} has no contemporaneous effect on the first variable. Under the maintained system, projecting $u_{d,t}$ on $u_{y,t}$ leaves $b_{22}\varepsilon_{2t}$. That innovation can be used in an LP, with its scale stated explicitly.

This design is credible only if the ordering and within-period timing are credible. With quarterly data, "no response within a quarter" can be much stronger than "the response takes a few minutes". The statistical Cholesky decomposition exists for any positive-definite covariance matrix; the economic interpretation of its columns requires the substantive exclusions.

5.1 Why some contemporaneous controls can be legitimate

Contemporaneous variables ordered before the policy shock may be controls under a valid recursive design because they are assumed unaffected by that shock on impact. This differs from indiscriminately controlling for variables measured after a policy action. For a total response, controlling for a mediator affected by the shock can remove part of the transmission mechanism or induce additional selection.

5.2 Changing the estimator does not change the identifying restriction

Plagborg-Møller and Wolf (2021) show the population equivalence of LP and VAR impulse-response estimands with unrestricted lag structures. Their results clarify how familiar structural identification schemes can be implemented with either approach. Finite lag truncation, regularisation and finite-sample precision remain important, but LP estimation is not a substitute for identifying the shock.

Other restrictions require equal care. Sign restrictions often yield a set of admissible responses rather than a unique response. Long-run restrictions constrain the cumulative or limiting response and require their own economic and time-series assumptions. Identification through changing volatility or non-Gaussianity relies on structural restrictions beyond a regression fit. Kolesár and Plagborg-Møller (2025) show why certain latent-shock identification schemes are particularly sensitive to nonlinear misspecification.

Teaching point. A recursive LP and a recursively identified VAR can share the same economic identifying assumption. The choice between their finite-sample estimators is a different decision.

Reading: Plagborg-Møller and Wolf (2021); Kolesár and Plagborg-Møller (2025).

6 Observed shocks: narrative and high-frequency designs

An observed series X_t can be used directly if the identifying argument supports its interpretation and the regression targets the desired response. A narrow event window or a narrative label is evidence about a design, not a mathematical guarantee of exogeneity.

6.1 Narrative information

Romer and Romer (2004) combine narrative records of intended policy changes with central-bank forecasts to remove systematic responses to anticipated conditions. The broader lesson is to explain which information drove the policy decision and why the remaining variation is unrelated to counterfactual outcomes. The completeness and timing of that information are substantive questions.

6.2 High-frequency surprises

Changes in market prices around announcements reduce the opportunity for unrelated news to intervene. They may nevertheless combine several components: current policy, forward guidance, information about the outlook, or other announcement news. A single surprise series need not identify all these components separately.

Bauer and Swanson (2023) document predictability of monetary surprises from information available before announcements and study adjustment for that information. Gertler and Karadi (2015) use high-frequency surprises as external instruments in a monetary VAR. These are useful examples of why the same type of measurement can enter different identification designs.

6.3 A shock measure is not always the structural shock itself

For a transparent measurement-error example, suppose

$$Y_{t+h} = \theta_h \varepsilon_t + u_{t+h}, \qquad X_t = \varepsilon_t + \nu_t,$$

where all cross-covariances among $\varepsilon_t, \nu_t, u_{t+h}$ vanish. Regressing Y_{t+h} directly on X_t gives

$$\beta_h^{\text{direct}} = \theta_h \frac{\text{Var}(\varepsilon_t)}{\text{Var}(\varepsilon_t) + \text{Var}(\nu_t)}.$$

The result is attenuated relative to the response per unit structural shock. Rescaling a noisy series by its own standard deviation does not remove this issue. In contrast, a valid proxy used as an instrument for a suitable policy variable can eliminate this particular attenuation in a ratio; Section 7 makes the required restrictions explicit.

6.4 Diagnostics that inform, but do not prove, validity

Check predictability from pre-announcement information, event timing, sensitivity to window choice, and unusual announcements. Distinguish an innovation in beliefs about future policy from unexpected current implementation. Inspect whether the series is being treated as a directly observed shock or a noisy proxy. Failure to reject a predictability test is not proof of the full causal exclusion restriction.

Teaching point. Document the economic source of the variation, its information content, measurement scale and intended use. A sophisticated estimator cannot resolve ambiguity about which shock the series measures.

7 External instruments and LP-IV

For a simple structural equation at horizon h, write

$$Y_{t+h} = \theta_h D_t + m_h(W) + u_{t+h}.$$

Let Q_t be an instrument and $\widetilde{Q}_t = Q_t - \mathbb{E}[Q_t \mid W]$. Sufficient moment conditions are

$$\mathbb{E}[\widetilde{Q}_t u_{t+h}] = 0, \qquad \mathbb{E}[\widetilde{Q}_t D_t] \neq 0.$$

They imply

$$\theta_h = \frac{\mathbb{E}[\widetilde{Q}_t Y_{t+h}]}{\mathbb{E}[\widetilde{Q}_t D_t]}. \tag{9}$$

The first condition is exclusion at the relevant horizon; the second is relevance. As with OLS, finite linear controls must correctly implement the adjustment needed for these moments. Under heterogeneous effects, this ratio needs a separate weighting interpretation. A general macroeconomic proxy-IV coefficient should not automatically be called a LATE.

7.1 A structural shock normalisation

Consider the zero-mean stationary linear representation

$$Y_t = \sum_{j\geq 0}\sum_k c_{jk}\varepsilon_{k,t-j}, \qquad D_t = \sum_{j\geq 0}\sum_k b_{jk}\varepsilon_{k,t-j}.$$

Suppose Q_t correlates with ε_{1t} and is orthogonal to every other dated structural shock entering the relevant outcomes, including nonzero leads and lags of shock 1. Then

$$\frac{\mathrm{Cov}(Q_t, Y_{t+h})}{\mathrm{Cov}(Q_t, D_t)} = \frac{c_{h1}}{b_{01}}, \qquad b_{01} \neq 0.$$

This is the response of Y per unit *impact response of* D to the identified shock. It includes subsequent endogenous policy reactions. It is not necessarily the effect of a policy path set externally to be one unit higher at all dates. A 25-basis-point normalisation multiplies this ratio by 0.25 if D is measured in percentage points.

7.2 Why dynamic exclusion is stronger than contemporaneous exclusion

The error in a horizon-h equation can contain earlier and later structural shocks. Orthogonality to other shocks only at date t need not eliminate those components. Stock and Watson (2018) establish conditions under which direct LP-IV can identify responses without invertibility. Their result requires a restrictive instrument condition. In their analysis, when lagged endogenous controls are needed to accommodate instrument correlations, validity can again require the conditions associated with invertibility. Thus "LP-IV never needs invertibility" is too broad.

Reading: Stock and Watson (2018). The display above is a deliberately strong, transparent sufficient-condition example, not the weakest version of their theorem.

8 Two IV examples and weak identification

8.1 A noisy proxy can work in a ratio

Let $D_t = b\varepsilon_t + v_t$, $Y_{t+h} = c_h\varepsilon_t + e_{t+h}$ and $Q_t = \varepsilon_t + \nu_t$. If the nuisance components are uncorrelated with Q_t and $b \neq 0$, then

$$\frac{\mathrm{Cov}(Q_t, Y_{t+h})}{\mathrm{Cov}(Q_t, D_t)} = \frac{c_h}{b}.$$

The noise ν_t may reduce precision without inducing the attenuation of a direct regression on Q_t. This conclusion fails if the proxy noise predicts the outcome error, loads on another relevant shock, or is correlated with the policy-variable nuisance v_t.

8.2 An instrument can be clean contemporaneously and fail dynamically

Take a scalar shock sequence ε_t with independent unit-variance entries and

$$Q_t = \varepsilon_t + a\varepsilon_{t-1}, \quad D_t = \varepsilon_t, \quad Y_t = \sum_{j\geq 0} c_j \varepsilon_{t-j}.$$

The instrument is relevant for the current shock, yet

$$\frac{\mathrm{Cov}(Q_t, Y_{t+h})}{\mathrm{Cov}(Q_t, D_t)} = c_h + ac_{h+1}.$$

The old shock also affects the future outcome. If ε_{t-1} is observed, conditioning on it removes this particular contamination. If it is latent, whether observable lagged variables suffice is a substantive information-recovery question. Increasing the number of lags without checking what they recover is not a proof.

8.3 Weak instruments concern information as well as precision

If $\mathrm{Cov}(Q_t, D_t) = 0$, the normalised response is not identified through this moment. If that covariance is small, conventional normal approximations for the ratio can be poor. Report first-stage evidence in the actual estimation sample and use inference designed for weak instruments when appropriate.

For one instrument and a candidate value θ_0, an Anderson–Rubin-style approach tests the moment

$$\mathbb{E}[\widetilde{Q}_t\{Y_{t+h} - \theta_0 D_t\}] = 0$$

without dividing by the estimated first stage. Inverting tests gives a confidence set, with the covariance calculation adapted to time dependence and the control specification. Such a set can be wide or unbounded. It still requires a valid instrument; weak-instrument robustness does not protect against exclusion failure.

With several endogenous shock components, the instrument–treatment covariance matrix must have sufficient rank. One proxy cannot generally disentangle a current-policy shock, a forward-guidance shock and an information shock without additional restrictions. State-specific IV additionally requires relevance and exclusion within the relevant state groups; unconditional validity does not automatically imply conditional validity.

9 Nonlinear responses: a causal slope is not every causal effect

Consider a directly observed, causally exogenous continuous shock X, independent of the other determinants of the potential outcome. Let

$$\Psi_h(x) = \mathbb{E}[Y_{t+h}(x)], \qquad \beta_h^{LP} = \frac{\mathrm{Cov}(X, \Psi_h(X))}{\mathrm{Var}(X)}.$$

Assume interval support (a, b), finite positive variance, differentiability and integrability sufficient for the following interchange of expectation and integration. Writing $\Psi_h(X) = \Psi_h(a) + \int_a^b \Psi_h'(x) 1\{X \geq x\}\, dx$ gives

$$\beta_h^{LP} = \int_a^b \omega(x) \Psi_h'(x)\, dx, \qquad \omega(x) = \frac{\mathbb{E}[(X - \mathbb{E}X) 1\{X \geq x\}]}{\mathrm{Var}(X)}. \tag{10}$$

For unbounded support, use a finite reference point and the corresponding tail conditions. The weight is nonnegative: it is proportional to the covariance of two increasing functions of X. Its integral is one, as can be checked by substituting $\Psi_h(x) = x$. Therefore the slope is a positively weighted average of marginal responses in this simple design. It need not equal the derivative at the mean, the usual unweighted average derivative, or a given finite contrast.

This calculation illustrates the observed-shock interpretation studied more generally by Kolesár and Plagborg-Møller (2025). Their proxy results require further conditions; the elementary observed-shock formula should not be transferred mechanically to arbitrary IV or misspecified control regressions.

9.1 Gaussian shocks and an exact cubic example

For $X \sim N(0, \sigma^2)$ and suitable regularity, Stein's identity gives $\beta_h^{LP} = \mathbb{E}[\Psi_h'(X)]$. For the teaching example $X \sim N(0, 1)$ and $\Psi_h(x) = g_h x + c_h x^3$,

$$\begin{aligned} \beta_h^{LP} &= g_h + 3c_h, \\ \Psi_h(\delta) - \Psi_h(0) &= g_h \delta + c_h \delta^3, \\ \mathbb{E}[\Psi_h(X + \delta) - \Psi_h(X)] &= \delta(g_h + 3c_h) + c_h \delta^3. \end{aligned}$$

All three formulas are causal under the example's assumptions. They answer different questions. At $g_h = 1, c_h = 1/4, \delta = 1$, the LP slope is 1.75, the fixed $0 \to 1$ effect is 1.25, and the realised-shift effect is 2.

Teaching point. Nonlinearity does not automatically destroy every causal interpretation of a linear LP. It changes which causal average is identified. To claim an exact finite-shift response, establish the additional restrictions that connect that average to the requested intervention.

Connection to policy relevance. Kitagawa, Wang and Xu (2025) discuss shape restrictions that can restore a nonnegative-weight representation for some estimands with negative weights, time variation in causal effects, and the connection between estimands and policy decisions. This is not a universal remedy for negative weights. It reinforces a further distinction: a well-defined causal average is not automatically the quantity needed for a particular policy choice.

10 State-dependent LPs: initial conditions and propagation

A common binary-state specification interacts the shock and controls with a pre-shock state S_{t-1}:

$$Y_{t+h} = \sum_{s=0}^{1} 1\{S_{t-1} = s\}\{\alpha_{hs} + \beta_{hs} X_t + \gamma_{hs}' W_{t-1}\} + u_{t+h}.$$

A continuous-state model may instead use $X_t g_h(Z_{t-1})$, approximated by splines. Full interaction of controls can matter when the nuisance function differs across regimes. A more flexible state function does not establish exogeneity or allow arbitrary nonlinearity in the shock.

10.1 Predetermined need not mean unrelated to past shocks

An initial state can depend on past policy and still be known before the current innovation. For example,

$$Y_t = \rho Y_{t-1} + a(Z_{t-1}) X_t + e_t.$$

Suppose current and future innovations have the required zero conditional means under the intervention, including the products $a(Z_{t+j-1})X_{t+j}$ at future dates. With constant propagation ρ, iterating conditional expectations gives

$$\mathbb{E}[Y_{t+h}(x) \mid \mathcal{F}_{t-1}] = \rho^{h+1} Y_{t-1} + \rho^h a(Z_{t-1}) x,$$

and the finite response is $\rho^h a(z)\delta$. The state can depend on earlier shocks. The validity of this example rests on its affine propagation and innovation restrictions, not merely on using a lagged state.

10.2 Shock-dependent transitions can change the response

For a contrasting example, let $Y_t(x) = x$, $Z_t(x) = 1\{x > c\}$ and $Y_{t+1}(x) = \rho(Z_t(x))x$. A change in x can cross the threshold and alter the propagation coefficient. Then

$$Y_{t+1}(x+\delta) - Y_{t+1}(x) = \rho(1\{x + \delta > c\})(x+\delta) - \rho(1\{x > c\})x.$$

It cannot generally be obtained by multiplying one fitted slope by δ. Conditioning on the initial state does not hold all future states fixed.

Gonçalves, Herrera, Kilian and Pesavento (2024) analyse the distinction between infinitesimal and finite responses in state-dependent systems. Their conclusions depend on the model and state mechanism they study. They should not be paraphrased as "every state affected by past shocks invalidates LP". The course manuscript gives a different sufficient-condition route: an affine conditional mean in the current shock with a coefficient determined by the initial state.

Teaching point. Ask whether the intervention changes the subsequent transmission mechanism, and whether the horizon-specific conditional mean remains affine in the shock. The word "endogenous" alone is too imprecise to answer this.

11 The identification argument in the course manuscript

The following restates the core logic of David, Giacomini, Jiao and Wang, *Causal State-Dependent Local Projections in Micro-Macro Data*, manuscript dated 8 November 2026. Let X_t be the common shock, $Z_{i,t-1}$ a pre-shock state, and $Y_{i,t+h}(x)$ the potential outcome.

Causal mean exogeneity. For every relevant x,

$$\mathbb{E}[Y_{i,t+h}(x) \mid \mathcal{F}_{t-1}, X_t] = \mathbb{E}[Y_{i,t+h}(x) \mid \mathcal{F}_{t-1}]. \tag{11}$$

Innovation condition and timing. $\mathbb{E}[X_t \mid \mathcal{F}_{t-1}] = 0$, and the initial state and predetermined controls are $\mathcal{F}_{t-1}$-measurable.

Affine conditional response. For each horizon,

$$\mathbb{E}[Y_{i,t+h} \mid \mathcal{F}_{t-1}, X_t] = g_h(Z_{i,t-1})X_t + r_{i,h}(\mathcal{F}_{t-1}). \tag{12}$$

Consistency, the appropriate support and integrability conditions, and positive conditional shock variance complete the argument. The affine restriction is on the conditional mean in the shock; g_h can be nonlinear in the state, and $r_{i,h}$ can be an unrestricted function of the past.

11.1 Step 1: obtain the causal interpretation

Consistency and (11) link the observed response surface to the counterfactual surface. By (12),

$$\mathbb{E}[Y_{i,t+h}(x) \mid \mathcal{F}_{t-1}] = g_h(Z_{i,t-1})x + r_{i,h}(\mathcal{F}_{t-1}).$$

Hence, over the support of the intervention,

$$\mathrm{IRF}_h(z;\delta) = \mathbb{E}[Y_{i,t+h}(X_t + \delta) - Y_{i,t+h}(X_t) \mid Z_{i,t-1} = z] = \delta g_h(z). \tag{13}$$

For continuous shocks this uses a suitably regular version of the conditional mean and the support requirements discussed in Section 2. The same affine condition makes fixed-value contrasts have the same slope.

11.2 Step 2: identify the coefficient from observed moments

Suppress the indices and write $Y = g_h(Z)X + r(\mathcal{F}_{t-1}) + u$, with $\mathbb{E}[u \mid \mathcal{F}_{t-1}, X] = 0$. Iterated expectations give

$$\begin{aligned}
\mathbb{E}[XY \mid Z] &= g_h(Z)\mathbb{E}[X^2 \mid Z] + \mathbb{E}[Xr(\mathcal{F}_{t-1}) \mid Z] + \mathbb{E}[Xu \mid Z], \\
\mathbb{E}[Xr(\mathcal{F}_{t-1}) \mid Z] &= \mathbb{E}[r(\mathcal{F}_{t-1})\mathbb{E}[X \mid \mathcal{F}_{t-1}] \mid Z] = 0, \\
\mathbb{E}[Xu \mid Z] &= \mathbb{E}[X\mathbb{E}[u \mid \mathcal{F}_{t-1}, X] \mid Z] = 0.
\end{aligned}$$

Therefore $g_h(z) = \mathbb{E}[XY \mid Z = z]/\mathbb{E}[X^2 \mid Z = z]$ when the denominator is positive. Step 2 is a statistical moment argument. Step 1 is needed to call the resulting function a causal response.

12 What the manuscript's conditions buy, and what they do not

12.1 An unrestricted past-information nuisance

For any integrable predetermined basis $\phi(Z)$,

$$\mathbb{E}[X_t \phi(Z_{i,t-1}) r_{i,h}(\mathcal{F}_{t-1})] = 0.$$

Thus the nuisance need not lie in the span of a small list of lag controls for this population moment to be valid. Predetermined controls can improve precision. A residual orthogonal to a finite collection of regressors need not satisfy $\mathbb{E}[X_t \mid \mathcal{F}_{t-1}] = 0$; replacing the shock by such a residual requires its own justification. Nor does the moment argument automatically justify finite-sample demeaning, regularisation, missing-data selection or a chosen confidence-band method.

12.2 Coarsening the state changes the target

Suppose the true coefficient is $\widetilde{g}_h(S)$ for a richer pre-shock state S, but the researcher reports only Z. With the innovation condition given the full history, define $v(S, Z) = \mathbb{E}[X_t^2 \mid S, Z]$. The coarse-state moment identifies

$$g_h^{LP}(z) = \frac{\mathbb{E}[v(S, Z)\widetilde{g}_h(S) \mid Z = z]}{\mathbb{E}[v(S, Z) \mid Z = z]}. \tag{14}$$

The unweighted finite-shift response divided by δ is $\mathbb{E}[\widetilde{g}_h(S) \mid Z = z]$. Their difference is

$$\frac{\mathrm{Cov}(v(S, Z), \widetilde{g}_h(S) \mid Z = z)}{\mathbb{E}[v(S, Z) \mid Z = z]}.$$

For example, two equally frequent subgroups with slopes 1 and 3 and shock variances 1 and 9 have unweighted slope 2 but LP slope $(1 + 27)/(1 + 9) = 2.8$. This example illustrates why similar state labels can conceal different population weights.

12.3 A linear state interaction is a further projection

If the true response is $g_h(Z)$, fitting $X_t(a_h + b_h Z)$ approximates the function using the shock-variance-weighted distribution of Z. Under constant conditional shock variance,

$$b_h = \frac{\mathrm{Cov}(Z, g_h(Z))}{\mathrm{Var}(Z)}.$$

For smooth functions and suitable support, this has a weighted-average-derivative interpretation in Z. It is not the response level. If Z is symmetric around zero and $g_h(Z) = 1 - Z^2$, then $b_h = 0$ despite substantial heterogeneity. With nonconstant variance, use the corresponding weighted distribution; ordinary covariance under the unweighted state distribution is no longer the formula.

Teaching point. The paper makes precise when a flexible state-response curve is an exact finite-shift response. It does not manufacture exogenous shocks, make every coarsened state sufficient, or turn nonlinear shock exposure into affine exposure.

13 Panels, common shocks and fixed effects

For a common shock X_t, a panel LP can use cross-sectional heterogeneity in state exposure:

$$Y_{i,t+h} = \alpha_{i,h} + \lambda_{t,h} + X_t g_h(Z_{i,t-1}) + u_{i,t+h}.$$

With unrestricted date effects, an arbitrary constant c_h can be transferred between the common response and the date effect:

$$g_h^*(z) = g_h(z) + c_h, \qquad \lambda_{t,h}^* = \lambda_{t,h} - c_h X_t.$$

The fitted outcome is unchanged. Therefore the overall level of the response curve is not identified from this specification alone. A reference-state normalisation, such as $g_h(z_0) = 0$, produces a relative response curve. Differences $g_h(z_A) - g_h(z_B)$ are invariant to that normalisation. They are causally interpretable only if the shock and state restrictions hold and there is adequate residual variation to identify the contrasts.

13.1 Date effects do not eliminate every common confounder

A common omitted variable with heterogeneous loadings, say $\ell_i C_t$, cannot generally be represented by one additive date effect. If X_t is correlated with C_t and ℓ_i is related to the state, the shock–state interaction can remain confounded. Unit effects remove time-invariant intercepts, not arbitrary time-varying confounding.

13.2 Unit transformations can change moment validity

If original regressors are predetermined but not strictly exogenous, subtracting a unit's time average mixes observations from different dates. Correlations that were absent in an original date-specific moment may appear after transformation. Dynamic-panel bias and the asymptotic framework therefore need separate attention. One should not infer validity of a fixed-effects estimator solely from the untransformed shock moment.

13.3 Common variation also changes uncertainty

Many firms exposed to the same monetary innovation do not supply as many independent shock realisations as firms times dates. A time-series covariance estimator based on date-aggregated scores can be appropriate under the course paper's assumptions. Cross-sectional dependence, overlapping horizons and serial correlation determine the covariance construction. These are inference issues distinct from the response-level normalisation and causal restrictions above.

For the course's firm-investment application, keep the distinction between absolute responses and state contrasts visible when presenting curves. Report the chosen baseline, shock units, horizon and sample. Do not describe a normalised curve's zero as an estimated absence of any total policy effect at that state.

Teaching point. Fixed effects implement particular comparisons. They do not serve as a universal identification argument, and date effects with common shocks can remove the very response level one might wish to report.

14 An applied identification workflow

1. **Write the target.** State the outcome, horizon, treatment or structural shock, intervention size, initial population and future policy rule. Distinguish a fixed contrast, a realised shift and a derivative.
2. **Write the timing.** List what agents and the econometrician know before the shock. Align announcements, observations, implementation and publication dates.
3. **Choose and defend the design.** Specify conditional exogeneity, recursive restrictions, observed-shock assumptions or an external-instrument design. Explain the economic reason for the restriction.
4. **Derive the population coefficient.** Apply residualisation or the IV moment. With heterogeneity or nonlinearity, calculate the implied weights rather than naming the coefficient an average effect by default.
5. **Connect the coefficient to the target.** State the affine restriction if multiplying a slope by a finite shock. For state dependence, address coarsening, state transitions and support.
6. **Check the estimable variation.** Assess collinearity, instrument rank, conditional support, state-specific first stages and fixed-effect normalisation.
7. **Choose estimation and inference.** Address lag approximation, nuisance estimation, serial and common-shock dependence, weak instruments, and the difference between pointwise and simultaneous uncertainty.
8. **Report diagnostics and limitations.** Present sensitivity to plausible control sets and timing choices, but explain which identifying assumptions remain untestable.

14.1 Common mistakes and the corresponding correction

Claim or practice	Correction
"LP is robust, so the coefficient is causal."	Robustness of estimation or inference does not establish shock exogeneity.
"The shock has no serial correlation."	Lack of serial correlation is weaker than a conditional innovation restriction, and neither excludes contemporaneous confounding.
"The state is lagged, so finite responses are identified."	Lagging fixes one timing problem; it does not establish the horizon-specific response restriction.
"Flexible splines solve identification."	Splines approximate a function of the state; they do not identify the shock or alter its causal meaning.
"Add all future controls."	Future variables can be mediators or colliders; their inclusion changes the comparison.
"A high-frequency surprise is automatically a valid IV."	Verify what it measures, relevance and the relevant dynamic exclusion restriction.
"A significant state interaction proves nonlinear policy effects."	Distinguish state heterogeneity from nonlinearity in shock size, and inspect the implied weights.
"Narrow bands validate the design."	Confidence statements are conditional on identification and approximation assumptions.

15 Worked checks for teaching and self-study

1. Serial correlation and an omitted lag. Suppose $X_t = \rho X_{t-1} + \eta_t$ is stationary and $Y_{t+h} = g_h X_t + \gamma_h X_{t-1} + u_{t+h}$, where $\mathbb{E}[X_t u_{t+h}] = 0$. What does a regression on X_t alone estimate?

Answer. Stationarity gives $\mathrm{Cov}(X_t, X_{t-1}) = \rho \,\mathrm{Var}(X_t)$, so the slope is $g_h + \rho\gamma_h$. Serial correlation creates bias here through an omitted relevant component. If $\gamma_h = 0$, serial correlation alone does not produce this bias. Adding the correct lag removes this particular statistical term, but does not prove that η_t is causally exogenous.

2. Misspecified control adjustment. Let $W \sim N(0,1)$ and independent zero-mean η have variance $\sigma_\eta^2 > 0$. Define $D = W^2 + \eta$ and $Y = \theta D + \lambda W^2 + u$, with $\mathbb{E}[u \mid D, W] = 0$. Does controlling linearly for W recover θ?

Answer. With an intercept, $P_W D = 1$ because $\mathrm{Cov}(W, W^2) = 0$. Hence $V^L = W^2 - 1 + \eta$ and

$$\beta^L = \theta + \frac{2\lambda}{2 + \sigma_\eta^2}.$$

Controlling for W^2 or correctly residualising by $\mathbb{E}[D \mid W] = W^2$ removes the term. The example separates conditional exogeneity from correct regression adjustment.

3. State coarsening. Two equally frequent states have causal slopes 1 and 3 and conditional shock variances 1 and 9. Compute the pooled slope.

Answer. It is 2.8, while the equal-population average slope is 2. Their difference comes from weighting, not estimation bias relative to the LP's own population target.

4. Finite shocks. Let $Y(x) = x + x^3/4$ and $X \sim N(0,1)$ exogenous. Interpret an LP coefficient of 1.75.

Answer. It equals the average derivative in this example. A fixed increase from 0 to 1 has effect 1.25; shifting the realised shock by 1 has average effect 2. Multiplying 1.75 by 1 gives neither finite response.

5. Date effects. If a common-shock panel includes unrestricted date effects and a flexible state interaction, can it identify the overall response level?

Answer. Not from that specification alone: adding a constant to the curve and subtracting the same constant times the shock from date effects changes no fitted values. A baseline convention identifies relative curves; state contrasts can remain identified under the other required restrictions.

6. A short empirical interpretation. Complete: "Under [design assumptions], the estimate measures [target and weighting] for [population], in response to [shock size and units], allowing [future adjustment], relative to [normalisation if any]." This sentence is a useful check before drawing a policy conclusion.

16 A selective reading guide

The references below organise the course's conceptual questions. Publication dates and versions are distinguished; the guide is selective, not a claim to list every paper on LP identification.

Starting point. Jordà (2005) introduces direct estimation by horizon. Jordà and Taylor (2025) provide a broad practitioner review. Read them to distinguish the estimation framework from the identifying design.

LP versus VAR. Plagborg-Møller and Wolf (2021) clarify population equivalence under unrestricted dynamics. This prevents attributing causal identification to the choice of estimator.

Potential outcomes. Rambachan and Shephard (2025 revision) formulate dynamic causal objects and conditions linking predictive estimands to those objects. This is useful before choosing a regression specification.

External instruments. Stock and Watson (2018) organise external-instrument identification, LP-IV and invertibility. Read the dynamic validity conditions, not just the one-equation IV formula.

Shock construction. Romer and Romer (2004), Gertler and Karadi (2015), and Bauer and Swanson (2023) illustrate narrative information, high-frequency instruments and the treatment of predictable announcement variation.

Control implementation. Lloyd and Manuel (2024) explain why two-step shock construction and one-step control adjustment need not be interchangeable outside the exact OLS equivalence case.

Nonlinear causal meaning. Kolesár and Plagborg-Møller (2025) study the causal interpretation of conventional estimands in nonlinear systems. The distinction between a weighted marginal effect and a finite intervention is central for teaching. Kitagawa, Wang and Xu (2025) discuss negative weights, time heterogeneity and policy relevance.

State dependence. Gonçalves et al. (2024) study state mechanisms and finite versus infinitesimal responses. David et al. (2026) supply the course's sufficient conditions for exact state-dependent finite-shift responses in micro–macro data. Compare the maintained models and targets rather than treating their conclusions as contradictory slogans.

Inference. Montiel Olea and Plagborg-Møller (2021) study LP inference. Keep their inference results separate from the causal assumptions needed to interpret a coefficient.

16.1 Suggested sequence for the short course

Use Sections 1–4 to establish the common identification logic, Sections 5–8 to compare designs, and Sections 9–12 to explain why nonlinear and state-dependent applications require care. Section 13 connects to the common-shock panel setting; Sections 14–15 can be used for participant discussion. The slides contain a shorter route, while these notes retain the derivations for follow-up reading.

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